

Molecular identification of wide-field amacrine cells in mouse retina that encode stimulus orientation

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Abstract

Summary

Visual information processing is sculpted by a diverse group of inhibitory interneurons in the retina called amacrine cells. Yet, for most of the >60 amacrine cell types, molecular identities and specialized functional attributes remain elusive. Here, we developed an intersectional genetic strategy to target a group of wide-field amacrine cells (WACs) in mouse retina that co-express the transcription factor *Bhlhe22* and the Kappa Opioid Receptor (KOR; B/K WACs). B/K WACs feature straight, unbranched dendrites spanning over 0.5 mm (~15° visual angle) and produce non-spiking responses to either light increments or decrements. Two-photon dendritic population imaging reveals Ca^{2+} signals tuned to the physical orientations of B/K WAC dendrites, signifying a robust structure-function alignment. B/K WACs establish divergent connections with multiple retinal neurons, including unexpected connections with non-orientation-tuned ganglion cells and bipolar cells. Our work sets the stage for future comprehensive investigations of the most enigmatic group of retinal neurons: WACs.

eLife assessment

Amacrine cells are a heterogeneous and understudied set of retinal interneurons. This study presents **valuable** new insights into the structure, function, and circuit connectivity of a particular subset of wide field amacrine cells (WACs). The authors use an impressive set of techniques to study structural and functional properties of these cells and to establish their postsynaptic circuit partners. Evidence for the central conclusions is **solid**, although some of the most interesting results could be pursued more completely.

Introduction

Inhibitory interneurons modulate information flow in the brain dependent on their diverse morphologies and synaptic mechanisms. Indeed, the remarkable diversity of inhibitory interneurons spans various brain regions, including the cortex, hippocampus, spinal cord, and retina (Booker & Vida, 2018 [DOI](#); He et al., 2016 [DOI](#); Sweeney et al., 2018 [DOI](#); Yan et al., 2020 [DOI](#)). A comprehensive understanding of the structure and function of inhibitory interneurons is crucial to decipher the functional dynamics and organizational principles of neural circuits.

In the retina, visual information is first encoded by photoreceptors, which transmit signals to the output neurons, the retinal ganglion cells (RGCs), via multiple, parallel excitatory pathways: photoreceptors → bipolar cells → RGCs (Baden et al., 2016 [DOI](#); Bae et al., 2018 [DOI](#); Euler et al., 2014 [DOI](#); Goetz et al., 2022 [DOI](#); Kerschensteiner, 2022 [DOI](#); Tran et al., 2019 [DOI](#)). Signaling in these pathways is modulated and shaped by the major class of inhibitory interneurons, known as amacrine cells, which are the most diverse yet least understood retinal cell class (MacNeil & Masland, 1998 [DOI](#); Masland, 2012 [DOI](#); Yan et al., 2020 [DOI](#)). Amacrine cells provide inhibitory synaptic input to bipolar cell terminals, RGC dendrites and other amacrine cells (Demb & Singer, 2015 [DOI](#); Diamond, 2017 [DOI](#)).

Classically, amacrine cell function is understood in the context of lateral inhibition that generates RGC receptive field surrounds (Farrow et al., 2013 [DOI](#); Flores-Herr et al., 2001 [DOI](#); Werblin, 1972 [DOI](#)). Amacrine cell functional diversity, however, extends well beyond this classical operation, as demonstrated by specialized computations in various cell types. For example, GABAergic synapses made by starburst amacrine cells are activated preferentially by specific directions of motion (Euler et al., 2002 [DOI](#); Jain et al., 2020 [DOI](#); Koren et al., 2017 [DOI](#)), which drives direction-selective (DS) firing in DS-RGCs (Briggman et al., 2011 [DOI](#); Lee et al., 2010 [DOI](#); Wei, 2018 [DOI](#)); VGlut3 amacrine cells are sensitive to local motion and provide excitatory input that generates object-motion selectivity in the downstream W3 RGCs (Kim et al., 2015 [DOI](#); Krishnaswamy et al., 2015 [DOI](#); Lee et al., 2014 [DOI](#)); S-cone amacrine cells are tuned to blue-ON signals, and their inhibitory output putatively drives blue-OFF responses in downstream RGCs (Chen & Li, 2012 [DOI](#); Sher & DeVries, 2012 [DOI](#)). Therefore, understanding how amacrine cells process visual features is imperative for deciphering retinal computation.

Systematic understanding of the diversity of amacrine cells and their functional roles relies on the ability to identify specific cell types and to interrogate systematically their roles in retinal circuits (Jo et al., 2018 [DOI](#); Zhu et al., 2014 [DOI](#)). Based on dendritic field diameters, amacrine cells can be divided into narrow-field (~30 – 150 μm), medium-field (~150 – 500 μm), and wide-field (>500 μm) subclasses (Masland, 2012 [DOI](#)). Among these, wide-field amacrine cells (WACs) are the least understood subclass. Stochastic labeling of WACs revealed dendrites that extend more than 0.5 mm from the soma, with a diameter of more than 1 mm (30° of visual angle) in mice, making them

the largest cells in the retina (Badea & Nathans, 2004 [↗](#); Lin & Masland, 2006 [↗](#); Perez De Sevilla Muller et al., 2007 [↗](#); Zhu et al., 2014 [↗](#)). Systematic explorations of their functions and synaptic connectivity, however, have been hindered due to the lack of genetic approaches for targeted functional recording and manipulation. In this study, we introduce an intersectional genetic approach to target a subset of non-spiking WACs featuring straight, unbranched dendrites. This approach enabled us to investigate both the functional properties and synaptic partnerships of these WACs, laying the foundation for future comprehensive studies of the roles these cells play in retinal information processing.

Results

Intersectional genetic approach for studying WACs

Our intersectional genetic approach to identify specific WACs combines three transgenic alleles: Cre and Flpo recombinases driven by two distinct genetic loci, and a reporter of dual (Cre- and Flpo-mediated) recombination (Cai, Kardon, et al., 2016; Jo et al., 2018 [↗](#)) (**Fig. 1A** [↗](#)). In our search for markers of WACs, the transcription factor *Bhlhe22* (also known as *Bhlhb5*) stood out. *Bhlhe22* has an essential role in the development of a spectrum of GABAergic amacrine cells as well as type 2 OFF cone bipolar cells (Feng et al., 2006 [↗](#); Huang et al., 2014 [↗](#)). Indeed, an existing transcriptomic database suggested that approximately half of GABAergic amacrine cells express mRNAs encoding *Bhlhe22* at a relatively high level (**Fig. 1B** [↗](#)). Given that WACs are primarily GABAergic (Wassle & Boycott, 1991 [↗](#); Zhu et al., 2014 [↗](#)), whereas narrow-field cells are primarily glycinergic (Wassle et al., 2009 [↗](#)), it is likely that unidentified *Bhlhe22*-expressing (*Bhlhe22*⁺) cells include uncharacterized WAC types. Indeed, the *Bhlhe22*-Flpo transgenic line, crossed with a Flpo-dependent reporter, labeled type 2 bipolar cells as well as amacrine cell processes that stratified at four levels of the inner plexiform layer (IPL) (Cai, Kardon, et al., 2016); and our pilot studies suggested that labeled amacrine cells include WACs.

To increase the genetic specificity in targeting WACs, we crossed the *Bhlhe22*-Flpo line with several Cre lines and discovered a compelling intersection with the Kappa Opioid Receptor (KOR)-Cre line, which labels amacrine cell processes that stratify primarily in two layers, including a subset of those labeled in the *Bhlhe22*-Flpo line (Cai, Huang, et al., 2016). Indeed, the *Bhlhe22*-Flpo x KOR-Cre intersection, combined with a dual ReaChR-mCitrine reporter line (Hooks et al., 2015 [↗](#)) (**Fig. 1A** [↗](#)), labeled a group of amacrine cells with processes that stratified primarily at the borders of the IPL (**Fig. 1C** [↗](#)) and very rarely labeled bipolar cells or RGCs. Furthermore, nearly all amacrine cells expressing the reporter (98%) also were labeled by *Bhlhe22* immunofluorescence (144/147 cells; n = 4 retinas) (**Fig. 1C-E** [↗](#)). For simplicity, we refer to the cells labeled by this intersectional genetic strategy as B/K amacrine cells.

Most (>80%) B/K cells had somas in the inner nuclear layer (INL), although some localized to either the ganglion cell layer (GCL) or the inner plexiform layer (IPL; **Fig. 1F** [↗](#)). The fluorescence profile of B/K processes, analyzed in stacks of confocal images, peaked in either ON or OFF layers, as defined by the known locations of ON or OFF bipolar cell axon terminals (Borghuis et al., 2013 [↗](#); Franke et al., 2017 [↗](#)) (**Fig. 1G** [↗](#)). Confocal images showed long, straight processes concentrated near the fluorescence peaks in ON or OFF regions of the IPL with sparse labeling in between, near the ON/OFF boundary (**Fig. 1H** [↗](#)).

B/K amacrine cells have long, unbranched processes and do not fire action potentials

To investigate the morphology and functional properties of single B/K amacrine cells, a cell in either the GCL or INL was targeted for whole-cell recording (Park et al., 2015 [↗](#); Park et al., 2020 [↗](#); Park et al., 2018 [↗](#)). The cell was filled with Lucifer Yellow through the pipette, and, after fixation,

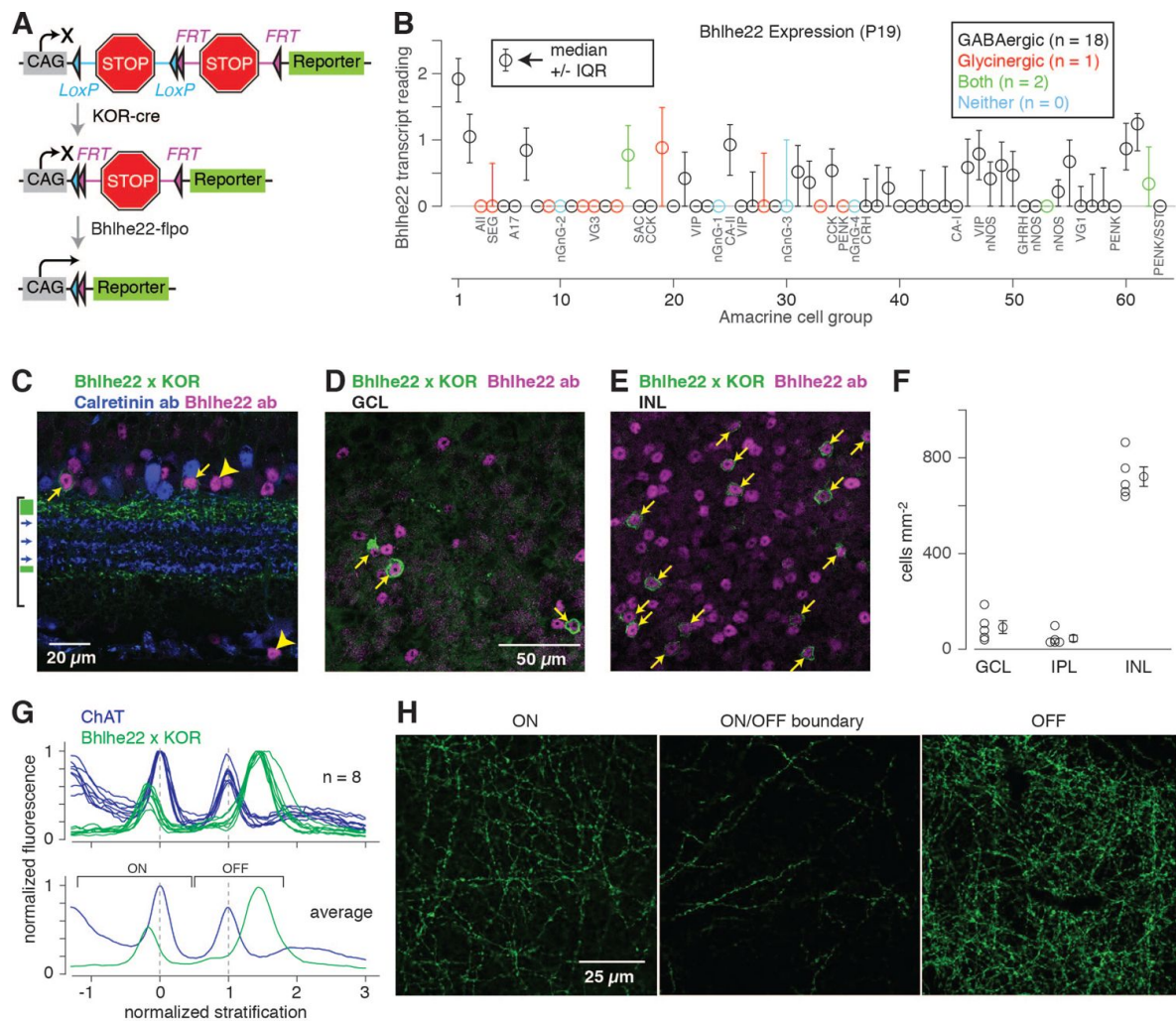


Figure 1.

Genetic strategies target *Bhlhe22*-expressing amacrine cells and cells that co-express *Bhlhe22* and KOR (B/K).

A. Intersectional genetic strategy. A KOR-Cre allele, acting on Loxp sites, combined with the *Bhlhe22*-Flpo allele, acting on Frt sites, removes two stop sequences and induces the expression of a reporter gene (either ReaChR-mCitrine fusion protein or tdTomato), driven by the CAG promoter.

B. *Bhlhe22* transcript readings in previously identified groups of amacrine cells (Yan et al., 2020). Data points show the median *Bhlhe22* transcript reading and the inter-quartile range (IQR) of these readings within each group. Cell groups are identified as GABAergic, glycinergic, both or neither, according to the definitions in the original study, and groups that correspond to known cell types are labeled with the name of the type (e.g., AII or A17 amacrine cell) or a characteristic gene name (e.g., VIP⁺ or nNOS⁺ amacrine cells).

C. Intersection between KOR-Cre and *Bhlhe22*-Flpo labels amacrine cell processes primarily in two layers (green rectangles, left) relative to calretinin bands (blue arrows; bracket marks the IPL). Top and bottom calretinin bands are identical to the OFF and ON ChAT bands, respectively. B/K amacrine cells were labeled with *Bhlhe22* antibody (yellow arrows); the antibody also labeled amacrine cells without reporter expression (yellow arrowheads).

D. Whole-mount retina showing B/K amacrine cells in the GCL with *Bhlhe22* antibody labeling (arrows).

E. Same as E. for the INL, which shows a higher density of labeled B/K amacrine cells in the GCL with *Bhlhe22* antibody labeling (arrows).

F. Density of B/K cells in different layers (n = 5 retinas) based on counts within 0.102 mm² regions. Error bars indicate $\pm$ SEM across samples.

G. Fluorescence profile of B/K intersectional genetic labeling relative to the ChAT bands in eight individual z-stacks (top). The average (bottom) shows fluorescence relative to the layers where ON and OFF bipolar cells stratify their axon terminals.

H. Example images of long, straight B/K amacrine cell processes in ON and OFF layers of the IPL and the boundary between these layers.

the dendrites were imaged by confocal microscopy (**Fig. 2A, B**). Dendrite stratification was normalized to the stratification of the ON and OFF ChAT (choline acetyltransferase) bands, which correspond to two sets of cholinergic starburst amacrine cell dendrites (**Fig. 2C**) (Manookin et al., 2008; Park et al., 2015; Park et al., 2020; Park et al., 2018). Low-resolution images of B/K WACs were traced to determine the number of extended dendrites, after initial branching within ~50 μm of the soma (**Fig. 2A, B**). In many cases, the entire dendrite could not be captured because the Lucifer Yellow signal faded or because of cuts in the tissue. Nevertheless, dendrites were commonly >0.5 mm in length, consistent with the general WAC morphology.

Dendrites of ON cells primarily stratified on the proximal side of the ON ChAT band, and those of OFF cells primarily stratified on the distal side of the OFF ChAT band; although four cells in the sample (1 ON, 3 OFF) stratified between the bands (**Fig. 2D**). For the entire sample ($n = 55$ cells), dendrite number versus stratification level showed two prominent clusters: OFF cells with ~4-6 dendrites and ON cells with ~7-13 dendrites (**Fig. 2D**). Stratification was measured within ~70-100 μm of the soma, but the dendrite retained its initial stratification level in distal dendrites, ~500 μm from the soma (**Fig. 2E**). Overall, B/K amacrine cells apparently comprise several WAC types with different stratification levels.

The stratification of amacrine cell dendrites relative to the terminals of ON and OFF bipolar cells in most cases predicts either ON or OFF responses to contrast modulation (Euler et al., 2014) (for exceptions, see Jo et al., 2023; Park et al., 2020). Indeed, B/K WACs with dendrites in the OFF layer depolarized during the negative/OFF phase of a contrast-reversing spot stimulus, whereas B/K WACs with dendrites in the ON layer depolarized during the positive/ON phase (**Fig. 3A**). For both ON and OFF B/K WACs, responses to increasing spot diameters showed increased modulation amplitude over a distance of ~1 mm (**Fig. 3B**), consistent with the diameter of their dendritic trees.

We did observe some heterogeneity in the B/K cells: the OFF B/K cells with somas in the GCL had relatively hyperpolarized resting potentials compared to the other groups (**Fig. 3C**); and OFF B/K WACs in the INL and GCL had similar patterns of response to contrast modulation, with relatively large depolarizing response to preferred contrast, whereas ON B/K cells had relatively large hyperpolarizing responses to non-preferred contrast (**Fig. 3D**). Notably, none of the recorded cells fired conventional action potentials (spikes). This differs from the response of another class of spiking WACs, which have distinct axonal and dendritic compartments (e.g., nNOS-1 and dopaminergic amacrine cells) (Badea & Nathans, 2004; Greschner et al., 2014; Kim et al., 2022; Lin & Masland, 2006; Murphy-Baum & Taylor, 2015; Newkirk et al., 2013; Park et al., 2020; Park et al., 2018; Stafford & Dacey, 1997; Taylor, 1996).

B/K WACs exhibit orientation-selective dendritic Ca^{2+} signals

The non-spiking nature of the B/K WACs suggests that they encode information through their dendrites. Further, their unique, highly oriented dendritic architecture evokes parallels with the elongated dendritic fields of a variety of orientation-selective (OS) and direction-selective (DS) neurons (Antinucci et al., 2016; Bloomfield, 1994; Kim et al., 2008; Kim et al., 2014; Nath & Schwartz, 2017; Weiler et al., 2022). To test whether B/K WACs encode orientation or direction information within their dendrites, we expressed the Ca^{2+} sensor GCaMP7s in B/K WACs using a recently developed dual reporter line (Ai195) and performed population dendritic two-photon imaging at multiple depths throughout the IPL while presenting gratings drifting in eight directions of motion. In both the ON and OFF layers, B/K dendrites exhibited substantial responses for two opposite directions of motion that shared the same orientation (**Figure 4A**). For individual regions of interest (ROIs), we computed both orientation- and direction-selective indices (OSI, DSI; see Methods). For the vast majority of ROIs, the OSI value was significantly larger than the DSI value (**Figure 4B**). ROIs with large OSI values were found at multiple levels of the IPL (**Figure 4C**).

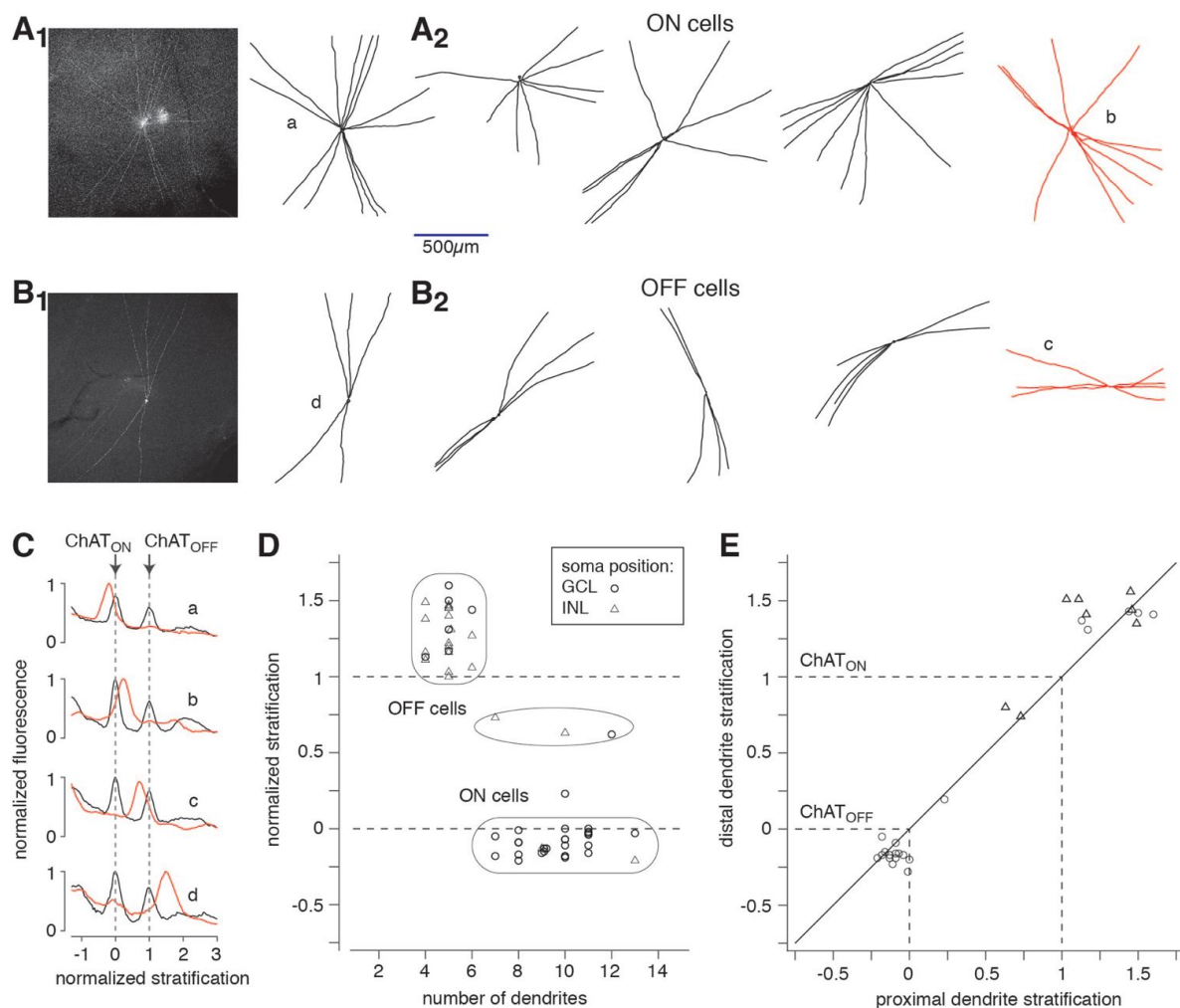


Figure 2.

B/K amacrine cells have long, unbranched dendrites and stratify at distinct levels of the inner plexiform layer.

A₁. Confocal image (left) and tracing (right) of a B/K wide-field amacrine cell (WAC) in the ON layer. **A₂.** Four additional ON B/K WACs. Cell at far right stratified between the ChAT bands. **B₁, B₂.** Same as A for OFF B/K WACs. Scale bar in A applies to all cells in A and B. Lower-case letters in A and B correspond to data in C. **C.** Fluorescence profiles of labeled dendrites of example ON and OFF WACs in A and B. Stratification is shown within each cell relative to ChAT bands labeled in the same tissue. Stratification is normalized to the inner/ON ChAT band (0, ChAT_{ON}) and the outer/OFF ChAT band (1, ChAT_{OFF}). **D.** Population analysis (n = 55 cells) showing the number of dendrites relative to the stratification, normalized to the ChAT bands. Apparent groups of cells are shown with outlines. Soma position is indicated by the symbol. **E.** A subset of B/K WAC dendrites were visualized near the soma (within ~150 μm) and at a distal location (~500 μm from the soma). Dendrites remained in their original layer, as indicated by the points falling near the identity line. Symbols indicate soma location, as in D.

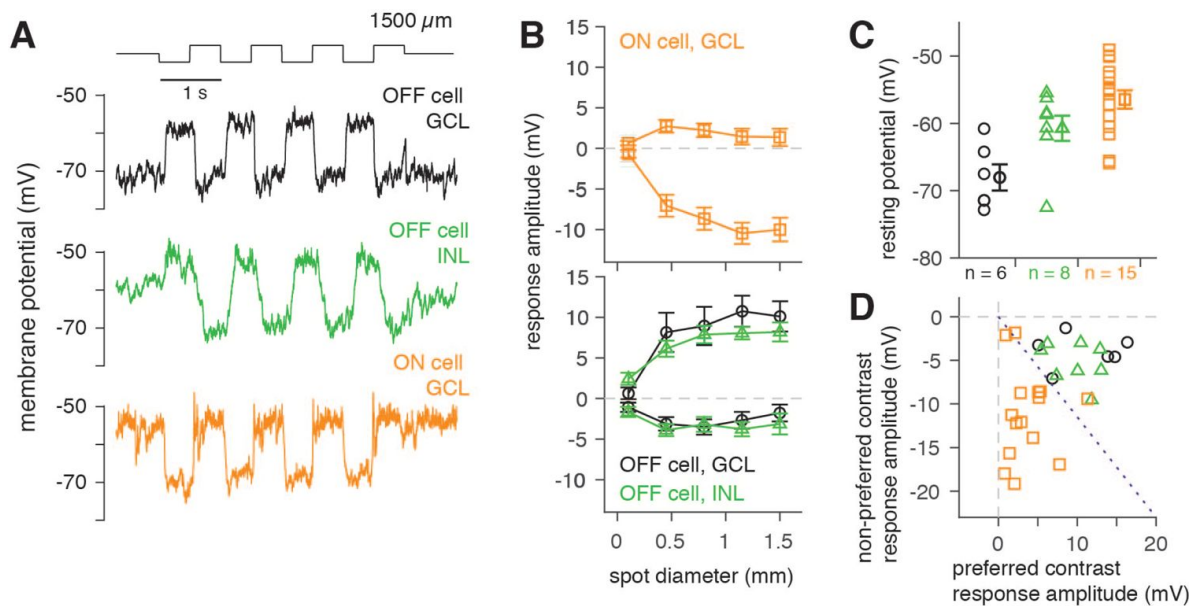


Figure 3.

B/K amacrine cells comprise ON and OFF types and do not fire action potentials.

A. Membrane potential recording of response to a contrast-reversing spot stimulus (1.5-mm diameter) for three B/K WACs, including two OFF cells with somas in either the GCL or INL and an ON cell with a soma in the GCL. **B.** Responses to contrast-modulated spots of varying diameters. Response amplitudes to the dark (OFF) and bright (ON) phases of the spot distinguish ON and OFF B/K WACs. Error bars in G. and I. indicate $\pm$ SEM across cells. The primary response (depolarization for OFF cells, hyperpolarization for ON cells) asymptotes at ~ 1 mm diameter. **C.** Resting membrane potential of B/K WACs in the presence of a mean luminance. OFF B/K WACs in the GCL were more hyperpolarized than OFF B/K WACs in the INL ($t(12) = 2.66$, $p = 0.02$) or ON B/K cells in the GCL ($t(19) = 4.64$, $p < 0.001$), whereas OFF B/K cells in the INL and ON B/K cells in the GCL were more similar ($t(21) = 1.84$, $p = 0.08$, two-sample t -tests). **D.** Peak depolarization to the preferred stimulus versus peak hyperpolarization to the non-preferred stimulus. Cells with equal modulation around the resting potential would fall along the dashed line. The combined group of OFF B/K WACs had larger depolarizing responses to preferred contrast compared to ON B/K WACs ($t(27) = 5.29$, $p < 0.001$), whereas ON B/K WACs had relatively larger hyperpolarizing responses to non-preferred contrast ($t(27) = 4.63$, $p < 0.001$).

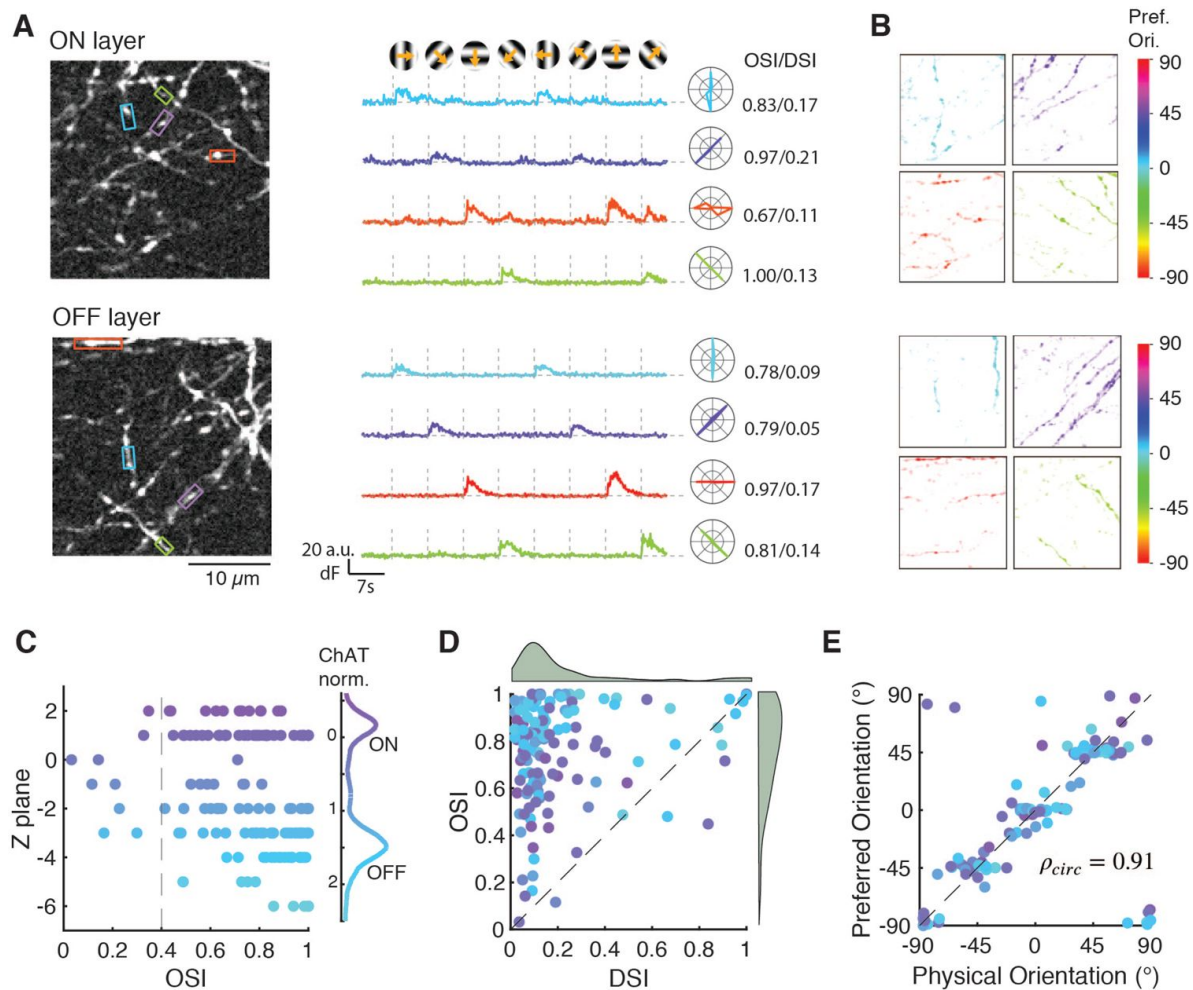


Figure 4.

Population dendritic imaging reveals orientation-selective B/K dendrites at different layers in the inner plexiform layer (IPL).

A. Example imaging planes for ON (top) and OFF (bottom) layers. Each plan shows a local correlation images over time. Example regions of interest (ROIs) are rectangles whose long axis aligns with the orientation of the dendrite. Corresponding baseline-subtracted fluorescence traces (in arbitrary units, a.u.) and polar plots of orientation tuning are shown on the right.

B. Spatial footprints tuned to four orientations extracted by sparse non-negative matrix decomposition for the exemplary ON (top) and OFF (bottom) layers shown in A. **C.** Orientation selectivity index (OSI) of dendritic ROIs (n = 135) at different IPL depths. **D.** OSI versus Direction-selective index (DSI) of B/K dendrites at different depths of the IPL. The smoothed probability density/kernel density of ROI OSI and DSI are plotted at the margins. OSI (0.75 ± 0.02 ; n = 135) is significantly larger than DSI (0.22 ± 0.02 ; $t(134) = 20.91$, $p < 0.001$). **E.** Physical orientation versus preferred orientation of B/K dendrites with OSI > 0.4 at different depths of the IPL (n = 126 ROIs). The circular Pearson correlation between physical and preferred orientations is 0.91 ($p < 0.001$).

Next, we evaluated the relationship between the preferred orientation tuning and the physical orientation of a dendrite. First, to visualize dendrites with common response properties from their overlapping neighbors, we utilized sparse non-negative matrix factorization (sNMF) for spatial footprint extraction from the time-lapse movies (Pnevmatikakis et al., 2016 [DOI](#); Wang & Zhang, 2013 [DOI](#)). This analysis revealed groups of straight, unbranched processes that resemble dendritic segments of filled B/K cells (**Figure 2A** [DOI](#)). Each group of processes exhibited a preferred functional orientation preference that closely aligned with their physical orientations (**Figure 4B** [DOI](#)). We next compared the physical orientation to the functional tuning of orientation-selective ROIs (i.e., those with OSI > 0.4) and found a strong structure-function correlation (**Figure 4E** [DOI](#)).

Collectively, we showed that B/K WAC dendrites encoded stimulus orientations that parallel their physical orientations. In theory, the orientation-selective Ca²⁺ responses could further drive orientation-tuned GABA release onto postsynaptic neurons.

B/K WACs make synapses with non-orientation tuned ganglion cells

B/K WAC dendrites stratify primarily at levels of the IPL that overlap the dendrites of commonly studied wide-field RGCs that do not encode orientation. Indeed, the OFF Delta cell (also known as OFF-Sustained Alpha RGC) (Krieger et al., 2017 [DOI](#); Margolis & Detwiler, 2007 [DOI](#); Tagawa et al., 1999 [DOI](#)) stratifies between the OFF ChAT band and the INL, overlapping the primary stratification of B/K WAC dendrites in the OFF layer (**Fig. 5A, B** [DOI](#)). The ON Alpha cell (also known as ON-Sustained Alpha RGC or the M4 melanopsin RGC) (Estevez et al., 2012 [DOI](#); Krieger et al., 2017 [DOI](#)) stratifies on the vitreal side of the ON ChAT band, overlapping the primary stratification of B/K WAC dendrites in the ON layer (**Fig. 5E, F** [DOI](#)). We tested whether B/K WACs make synapses with these RGC types using optogenetics.

After blocking transmission from photoreceptors (Park et al., 2015 [DOI](#); Park et al., 2020 [DOI](#); Park et al., 2018 [DOI](#); Pottackal et al., 2020 [DOI](#); Pottackal, Singer, & Demb, 2021 [DOI](#); Pottackal, Walsh, et al., 2021), light pulses that activated ReaChR expressed in B/K WACs evoked inhibitory postsynaptic currents (IPSCs) in RGCs (see Methods). IPSCs in OFF Delta RGCs were substantial (133 ± 46 pA; $n = 11$) and blocked largely by the GABA-A receptor antagonist gabazine (50 μ M SR95531) (**Fig. 5C** [DOI](#)); addition of glycine receptor antagonist strychnine (1 μ M) eliminated the IPSCs completely (**Fig. 5C** [DOI](#)).

Additionally, the evoked IPSCs increased monotonically with the intensity of light stimulation and ReaChR activation (**Fig. 5D** [DOI](#)). Similar results were observed in recordings from ON Alpha RGCs: IPSCs (231 ± 32 pA; $n = 19$) were blocked largely by gabazine and completely blocked by addition of strychnine (**Fig. 5E-H** [DOI](#)).

In comparison to the IPSCs recorded in OFF Delta and ON Alpha RGCs, much smaller responses were observed in the OFF-Transient Alpha cell (28 ± 11 pA; $n = 8$), which stratifies between the ChAT bands and overlaps B/K WAC processes more sparsely (**Suppl. Fig. 1A-C** [DOI](#)), and in DS-RGCs (18 ± 8 pA; $n = 4$) that co-stratify with the ChAT bands (**Suppl. Fig. 1D**). Thus, B/K WACs provide particularly robust synaptic input to OFF Delta and ON Alpha RGCs.

B/K WACs make synapses with non-orientation tuned bipolar cells in the OFF layer and implement sign-inverted feedforward inhibition (SIFI)

To probe possible connections between B/K WAC dendrites and bipolar cells, we performed connectomic analyses. We started with an existing dataset, SBEM (scanning block-face electron microscopy) volume k0725 (Ding et al., 2016 [DOI](#)), in which all of the inhibitory synapses onto an OFF Delta RGC had been annotated and some presynaptic amacrine cells, including WACs, had

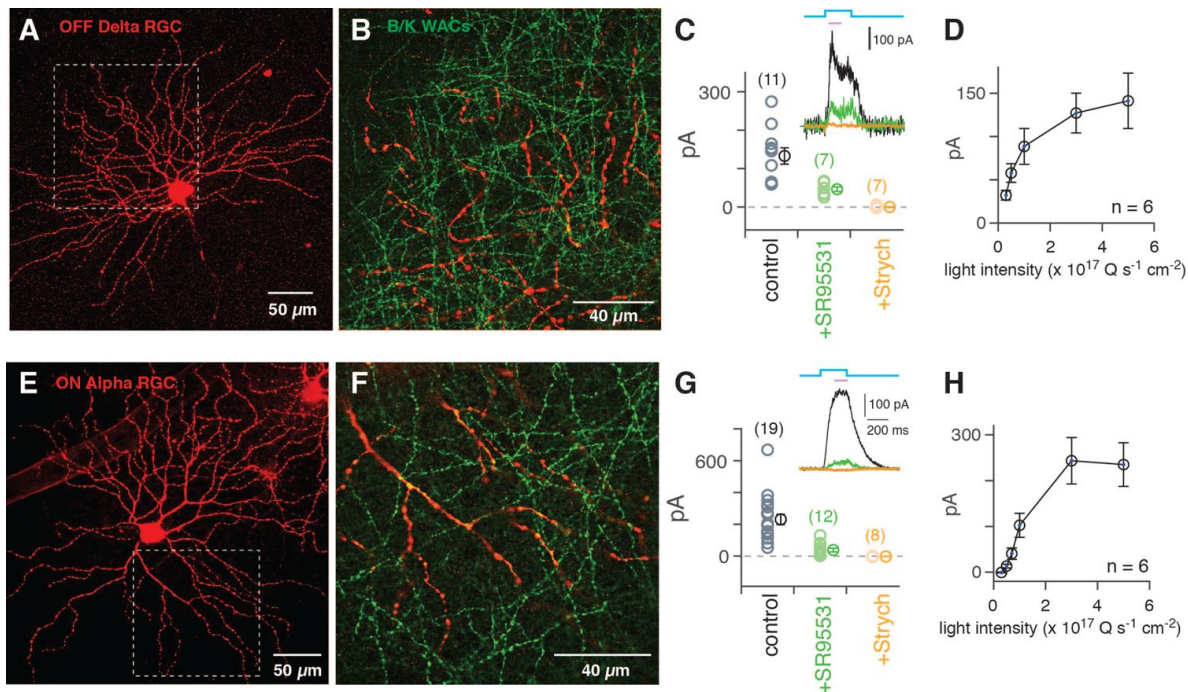


Figure 5.

B/K amacrine cells make synapses with non-orientation tuned alpha/delta-type RGCs.

A. Dendritic tree of a recorded OFF Delta RGC (also known as a Sustained OFF Alpha RGC). Image shows average fluorescence in a confocal stack. Dashed square shows region in B. The RGC was filled with Lucifer Yellow (LY) during whole-cell recording, which was subsequently amplified with LY primary antibody and a red secondary antibody. **B.** Single confocal section showing OFF Delta RGC dendrites from A. relative to B/K WAC dendrites, labeled by the Cre/Flpo-dependent ReaChR-mCitrine reporter. **C.** Optogenetic stimulation ($5.3 \times 10^{17} \text{ Q s}^{-1} \text{ cm}^{-2}$) of ReaChR-expressing B/K WACs caused an inhibitory postsynaptic current (IPSC, black) in an OFF Delta RGC. The IPSC ($133 \pm 46 \text{ pA}$; $n = 11$) was mostly blocked by gabazine ($50 \mu\text{M}$ SR95531; $46 \pm 6 \text{ pA}$; $n = 7$; difference of $106 \pm 25 \text{ pA}$, $t(6) = 4.22$, $p = 0.0056$) and completely blocked by subsequent addition of strychnine ($1 \mu\text{M}$; $-0.33 \pm 1.17 \text{ pA}$; $n = 7$; difference of $46 \pm 5 \text{ pA}$, $t(6) = 8.68$, $p < 0.001$). Error bars in all panels indicate $\pm$ SEM across cells. Optogenetic experiments were performed in the presence of glutamate receptor blockers (see Methods) to prevent photoreceptor contributions to the light response. **D.** Increasing optogenetic stimulation caused increasing IPSC amplitude in OFF Delta RGCs. **E-H.** Same as A-D. for ON Alpha RGCs (also known as Sustained ON Alpha RGCs). The ON Alpha RGC IPSC ($231 \pm 32 \text{ pA}$; $n = 19$) was mostly blocked by gabazine ($38 \pm 12 \text{ pA}$; $n = 12$; difference of $214 \pm 41 \text{ pA}$, $t(11) = 5.23$, $p < 0.001$) and completely blocked by strychnine ($-2.9 \pm 0.8 \text{ pA}$, $n = 8$; difference of $58 \pm 14 \text{ pA}$, $t(7) = 4.24$; $p = 0.0038$).

been traced (Grimes et al., 2022 [↗](#)) (**Fig. 6A-C** [↗](#)). The OFF Delta RGC in k0725 received 1298 inhibitory (non-ribbon) synapses, with ~23% of these (278) coming from the narrow-field, glycinergic AII amacrine cell (Demb & Singer, 2012 [↗](#)). Of the remaining synapses, 150 amacrine cells were reconstructed at least partially (Grimes et al., 2022 [↗](#)), and these provided a total of 300 (out of 1020 non-AII) synapses to the OFF Delta RGC. This group comprised three classes: bistratified amacrine cells with dendrites in at the borders of the INL and GCL (Sublaminae S1 and S5) (**Fig. 6B** [↗](#)); amacrine cells with asymmetric dendritic trees (**Fig. 6B** [↗](#)); and single WAC neurites that ran unbranched through S1 of the IPL, in the same plane as the dendrites of the OFF Delta RGCs (**Fig. 6C** [↗](#)).

We quantified properties of the putative B/K WAC neurites that made synapses with OFF Delta RGCs by evaluating synapse density and organization of nearby output synapses from straight, unbranched neurites. There were apparently two groups of WAC neurite with either sparse or dense total number of synapses (combined inputs and outputs) (**Fig. 6C** [↗](#)). At least some of these straight neurites might belong to B/K WACs and we therefore sought to address this hypothesis to gain information about the synaptic connectivity of these cells at the ultrastructural level.

Additionally, we found that ~25% of the inhibitory synapses to the OFF Delta RGC arose from an amacrine cell element that also provided an inhibitory synapse to a presynaptic type 2 cone bipolar cell that itself was presynaptic to the OFF Delta RGC; thus, there appeared to be simultaneous pre- and postsynaptic inhibition arising at a substantial number of sites on the OFF Delta RGC dendrite. We term this synaptic motif **SIFI**, for **Sign-Inverting Feedforward Inhibition** (**Fig. 7C** [↗](#)). SIFI is found also in AII synaptic output (140/278; ~50% of synapses), which is expected given that AIIs provide inhibitory input both to OFF Delta RGCs and to type 2 OFF cone bipolar cells presynaptic to these RGCs (Graydon et al., 2018 [↗](#); Tsukamoto & Omi, 2017 [↗](#)). Notably, synapses from other amacrine types also exhibit SIFI (205/1020; ~20%) with the OFF Delta RGC. By contrast, the SIFI motif was observed infrequently at a reconstructed OFF Alpha RGC, where only ~8% of AII amacrine cell synapses (20/248) and ~4% of non-AII amacrine cells (36/916) exhibited SIFI (data not shown). Thus, SIFI seems to be implemented preferentially at synapses onto the OFF Delta RGC as compared to the OFF Alpha RGC and includes synapses made by putative B/K WACs.

Because amacrine cell types cannot be identified from the SBEM volume based on morphologies alone, we performed ultrastructural analysis of a new volume (**Suppl. Fig. 2A**) in which B/K WACs expressed a genetically-encoded marker for use with SBEM. We targeted the enhanced peroxidase dimeric APEX2 (Joesch et al., 2016 [↗](#); Zhang et al., 2019 [↗](#)) to the mitochondrial matrix of B/K WACs by dual Cre/Flp-mediated recombination and visualized genetically-identified neurites by the DAB reaction (**Fig. 6D-G** [↗](#)). Assessment of APEX2 expression revealed DAB⁺ mitochondria in varicosities that are presynaptic to presumed RGC dendrites (i.e., processes that lack synaptic vesicles) and OFF cone bipolar cell terminals in S1 (i.e., axon terminals that contained ribbon synapses and vesicles) (**Fig. 6G** [↗](#)). Furthermore, some of these B/K WAC processes exhibited SIFI-connectivity (**Fig. 6G** [↗](#)), matching the pattern of straight WAC neurites reconstructed from the k0725 SBEM volume. The labeled B/K WAC dendrites exhibited a relatively high density of synapses, matching the subset of WAC neurites with high synapse density reconstructed from k0725 (**Fig. 6E** [↗](#); **Suppl. Fig. 2B, C** [↗](#)) and include outputs to cone bipolar cells (**Fig. 6E, F** [↗](#)). The combined analysis of optogenetic and connectomic data suggest that B/K WACs, which carry orientation-tuned signals (**Fig. 4** [↗](#)), make output synapses to postsynaptic cells that are not themselves orientation selective.

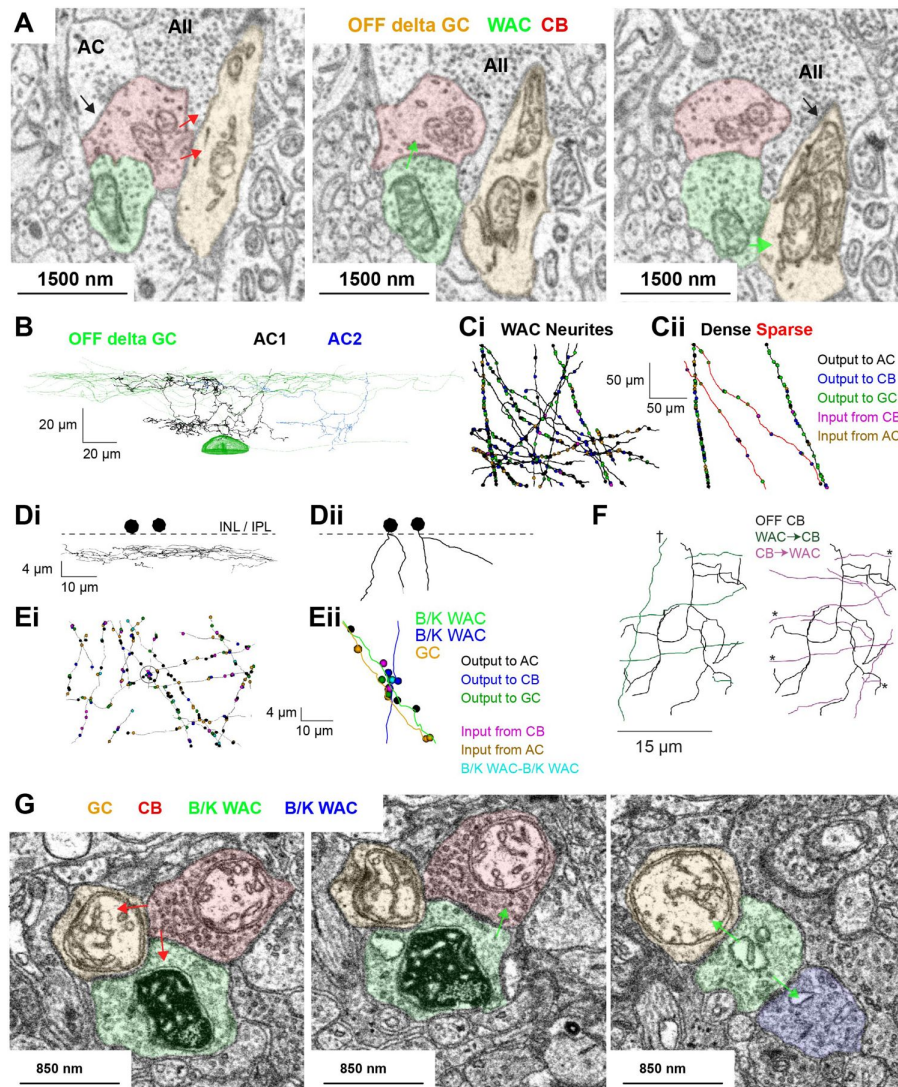


Figure 6

Connectomic analysis uncovers synaptic connections between B/K WACs and bipolar cells.

A. SBEM sections from dataset k0725 show connectivity between a WAC (green), an OFF cone bipolar cell terminal (CB; red), and an OFF Delta RGC (yellow). Also shown are an unidentified amacrine cell (AC) and an AII amacrine cell (AC). Cell-type identification was determined from reconstructions in a previous study (Grimes et al., 2022). The WAC makes synapses (green arrows, middle and right panels) onto both an OFF bipolar cell terminal and the RGC dendrite that is postsynaptic to the OFF bipolar cell terminal, the signature of sign-inverted feedforward inhibition (SIFI). Additional BC synapses (red arrows) and non-WAC AC synapses (black arrows) are indicated. **B.** OFF Delta RGC with two amacrine cell types that also make SIFI synapses. Reconstructions (side view) are from a previous study (Grimes et al., 2022). **Ci.** Sample WAC neurites (top-down view) with SIFI synapses. Contacts are color-coded per the text key. **Cii.** Some WACs (black lines) exhibit dense synaptic connectivity; whereas others (red lines) have sparse connectivity. **Di.** Side view of the reconstruction of DAB⁺ processes in the OFF layer adjacent to the INL (S1) of a B/K dAPEX2 retina SBEM dataset. Somas (black circles) demarcate the INL/IPL boundary. **Dii.** Labeled somas descended beyond the volume towards the ON layer; none of the reconstructed dendrites in the OFF layer were connected to a soma within this small volume. **Ei.** Sample of reconstructed B/K WAC neurites (top-down view) with synapses color-coded per text key. **Eii.** Connections between two B/K WACs (green and blue dendrites) and an RGC (orange) from circled area in Ei. **F.** Partial reconstruction of a cone bipolar (CB) cell terminal with WAC dendrites that make inputs (green) and receive outputs (magenta) from the CB. Some dendrites are only presynaptic to the CB (†), whereas others are both pre- and postsynaptic (*). **G.** SBEM: left, OFF cone bipolar (CB, red) ribbon-type synapse (red arrows) to RGC (orange) and to DAB⁺ WAC (green). Center, DAB⁺ WAC inhibitory feedback synapse to CB (green arrow). Right, inhibitory synapses (green arrows) from DAB⁺ WAC to RGC and to another DAB⁺ WAC.

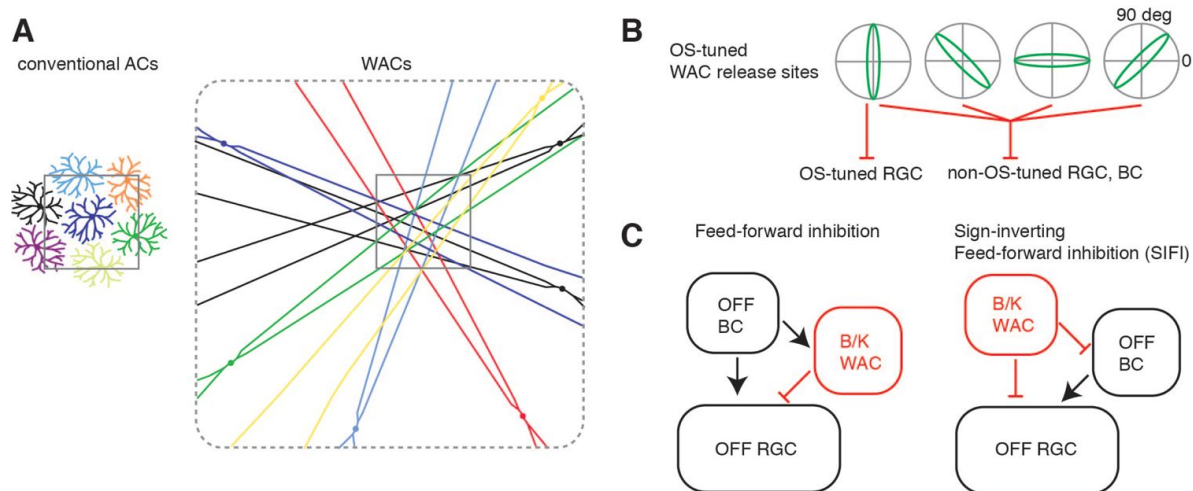


Figure 7.

Proposed circuit function of B/K WACs

A. A mosaic of amacrine cells can provide coverage to a region of retina (square) with either tiled processes of seven narrow-field cells or straight dendrites projecting from the same number of WACs. **B.** Either selective or non-selective integration of OS-tuned WAC dendrites can result in either OS-tuned or non-OS-tuned postsynaptic neurons. **C.** Schematic diagrams for feed-forward inhibition and sign-inverted feed-forward inhibition (SIFI).

Discussion

We developed an approach for systematic study of an enigmatic group of retinal interneurons: non-spiking wide-field amacrine cells (WACs) with straight, unbranched dendrites that are either ON or OFF cells, consistent with their stratification level in the IPL (**Figs. 1–3**). Ca^{2+} imaging showed that B/K WAC dendrites exhibited functional orientation tuning, consistent with their physical orientation (**Fig. 4**). Optogenetic studies showed that B/K WACs made output synapses onto non-orientation tuned RGC types, driving strong GABAergic inhibitory currents in ON Alpha and OFF Delta RGCs (**Fig. 5**). Connectomic analyses showed that B/K WACs make synapses with bipolar cell axon terminals, which likewise lack orientation tuning (**Fig. 6**) (Matsumoto et al., 2021). The nature of encoding visual images with orientation-tuned amacrine cell dendrites suggests a novel organization for WACs. Rather than encoding local subregions via a tiled mosaic of narrow-field cells, WACs apparently overlap substantially with their neighbors and cover orientation space through the collective processing across cells with different dendrite orientations (**Fig. 7A**). Networks of dendrites that are individually orientation-tuned could deliver a net tuned or non-tuned inhibition to postsynaptic neurons, depending on the pattern of convergence (**Fig. 7B**). The connections shown here suggest that orientation-tuned inhibition plays an unexpected role in controlling the responses of non-orientation-tuned postsynaptic neurons in the retina (**Figs. 5, 6**).

WACs form a diverse group of interneuron types in mammalian retina

B/K WACs share some properties with previously described cells visualized with methods for sparse labeling. Two studies identified a total of six WAC types that lacked separate dendritic and axonal compartments and can be compared to the B/K WACs: Cluster 1 and bifid cells (Badea & Nathans, 2004); and WA1, WA2-2, WA3-2 and WA4-3 cells (Lin & Masland, 2006). In comparing the soma location and dendritic tree properties, only bifid cells have properties consistent with those of B/K WACs: relatively asymmetric dendritic trees and mainly in the OFF layer with somas in the INL (Badea & Nathans, 2004). Collectively, these results suggest a diversity of WAC types in the mouse retina that extends beyond the B/K WACs described here.

Additional evidence for WAC diversity comes from both anatomical and physiological studies in mouse and other species. For example, SBEM reconstruction of mouse retina identified two groups of straight, unbranched dendrites that made synapses with one of two types of ON bipolar cell terminals (either Type 7 or Type 5A) (Hanson et al., 2023). Further, calcium signals in Type 5A bipolar cells showed orientation tuning, particularly for vertical orientations (Hanson et al., 2023). The working model for this circuit presumes that straight WAC dendrites produce orientation-tuned signals and convey this information to the Type 5A bipolar cell terminals via gap junctions; and these orientation-tuned signals were apparently further conveyed to postsynaptic DS-RGCs via glutamate synapses. The presumed orientation-tuned signals in WAC dendrites are supported by the direct evidence of OS tuning shown here (**Fig. 4**). However, the B/K WACs apparently comprise different cell types than those connecting Type 5A bipolar cells, which would presumably stratify proximal to the ON ChAT band and between the ChAT bands (Hanson et al., 2023).

WACs with similar morphology to B/K cells were described as ‘wiry’ in the primate retina (human and monkey) and stratify in at least three layers: ON, OFF and near the ON/OFF boundary (Kolb et al., 1992; Manookin et al., 2015; Mariani, 1990). Similar WACs were identified in rabbit retina as well as in other species (MacNeil et al., 1999). The collected findings suggest that WACs with long, straight dendrites exist across species and across multiple layers of the retina within a species. To the extent that WACs with long, straight dendrites commonly convey orientation-tuned GABA release, this property must be fundamental across mammalian retina. Notably, OS signals

are also conveyed by at least one type of spiking, axon-bearing WAC in rabbit (Murphy-Baum & Taylor, 2015 [↗](#)), and they are also present within a narrow-field non-spiking glycinergic amacrine cell in mouse (Mani et al., 2023 [↗](#)).

Synaptic properties of B/K WACS

B/K WACs appear to be GABAergic based on the optogenetic response in postsynaptic RGCs, which was primarily blocked by a GABA-A receptor antagonist. But, some B/K WACs may have mixed GABA/glycine release (Lu et al., 2008 [↗](#); Sawant et al., 2021 [↗](#)) (Fig. 1B [↗](#)), given that residual IPSCs were blocked by the glycine receptor antagonist strychnine. This was not the case in studies of other amacrine cell types performed in other Cre lines (VIP, CRH, ChAT nNOS); in these lines, presumed GABAergic synapses, activated by stimulation of ChR2, were blocked completely by a GABA-A receptor antagonist (Park et al., 2015 [↗](#); Park et al., 2020 [↗](#); Park et al., 2018 [↗](#); Yonehara et al., 2011 [↗](#)). An alternate explanation for strychnine-sensitive responses evoked by the optogenetic stimulation of B/K cells is polysynaptic activity, e.g., B/K WACs could disinhibit glycinergic cells or could excite glycinergic cells via gap junctions (Jo et al., 2023 [↗](#)). Further studies would be required to investigate these possibilities.

The reconstruction of B/K dendrites showed a prominent pattern of SIFI, which would presumably produce a powerful form of inhibition that hyperpolarizes the RGC dendrite while simultaneously suppressing glutamate release from the presynaptic bipolar cell terminal. The SIFI synaptic motif of B/K WACs resembles the powerful glycinergic inhibition mediated by the AII amacrine cell (Demb & Singer, 2012 [↗](#); Ke et al., 2014 [↗](#)), and in both cases it seems to be a specific property of input to the OFF Delta RGC but not to the OFF Alpha RGC.

Possible roles of B/K WACs in retinal information processing

Amacrine cells with strong feature tuning to direction, motion or color are presumed to generate a similar degree of tuning in postsynaptic RGCs (Chen & Li, 2012 [↗](#); Sher & DeVries, 2012 [↗](#)). For example, starburst amacrine cell pre-synapses activated by leftward motion would shape the responses of postsynaptic DS RGCs that report rightward motion (Briggman et al., 2011 [↗](#); Wei, 2018 [↗](#)). Similar circuits mediate OS tuning in some RGC types in both mouse and zebrafish, where an RGC tuned to one orientation (e.g., horizontal) would receive inhibition tuned to the orthogonal orientation (e.g., vertical) (Antinucci et al., 2016 [↗](#); Nath & Schwartz, 2016 [↗](#); Venkataramani & Taylor, 2016 [↗](#)). Electrical coupling to OS-tuned WAC dendrites also can drive OS tuning in postsynaptic bipolar cells and RGCs (Hanson et al., 2023 [↗](#); Nath & Schwartz, 2017 [↗](#)).

We have shown here an additional pattern whereby OS-tuned amacrine cells could influence RGC and OFF bipolar cells that are not themselves OS-tuned. Presumably, the postsynaptic RGC or bipolar cell collects inhibitory synapses from B/K WACs of multiple orientations, preventing a strong net OS tuning in the population of connected dendrites and ensuring a lack of OS-tuning in the postsynaptic response (Fig. 7A, B [↗](#)). There may be an advantage to encoding visual images with OS-tuned interneurons (Fig. 7A [↗](#)), for example, since oriented features form core components of natural images (Olshausen & Field, 1996 [↗](#)). The oriented dendrites of a B/K WAC could also promote the correlation of signaling across distant points on the retina (McIlwain, 1964 [↗](#); Neuenschwander & Singer, 1996 [↗](#)).

There are other cases where tuned neurotransmitter release from presynaptic sites is canceled in the postsynaptic neuron. For example, acetylcholine released from starburst amacrine cells onto RGCs lacks strong DS tuning, because cholinergic transmission is paracrine and converges across multiple angles of direction tuning (Lee et al., 2010 [↗](#); Pottackal et al., 2020 [↗](#); Sethuramanujam et al., 2021 [↗](#)). A similar model explains how OS-tuned glycinergic release from narrow-field VGLUT3 amacrine cells could produce non-OS-tuned glycinergic inhibition of ON DS-RGCs (Mani et al., 2023 [↗](#)).

Limitations of the study and next steps in understanding WAC function

Our genetic approach to labeling B/K WACs has obvious advantages for targeting cells systematically. Still, though, it remains to be understood how many cell types and what fraction of an individual type is identified in any given mouse line. For example, do the B/K WACs that stratify between the ChAT bands represent a rare and sparse cell type, or are we observing sparse labeling of a dense cell type? Our inquiries therefore would benefit from additional functional tests that distinguish B/K WAC types based on Ca^{2+} imaging or by molecular markers that label soma mosaics.

In the future, we also seek methods for inactivating B/K WAC types. Inactivation studies could evaluate potential roles for B/K WACs in regulating overall excitability of RGCs and shaping OS tuning in specific RGC types. Further, there is a possible role that WACs could play in development of other retinal cell types via temporary connections during postnatal development (Gamlin et al., 2020 [↗](#)). By controlling B/K WAC activity more specifically, one could also learn how these cells shape responses that influence activity in the brain and behavior (Cruz-Martin et al., 2014 [↗](#); Hillier et al., 2017 [↗](#); Schroder et al., 2020 [↗](#)).

Methods

Experimental model and subject details

All animal procedures were approved by the Institutional Animal Care and Use Committees at Yale University, University of Maryland or University of Pittsburgh and were in compliance with National Institutes of Health guidelines. For *Bhlhe22*-Flpo knock-in/knock-out mice, Flpo recombinase was targeted to the endogenous *Bhlhe22* (*Bhlhb5*) locus, where it replaced the *Bhlhe22* gene (Cai, Kardon, et al., 2016). For KOR-Cre knock-in/knock-out mice, Cre recombinase was targeted to the endogenous KOR locus (*Oprk1*) (Cai, Huang, et al., 2016) (Jackson Laboratory #035045). Dual Cre/Flpo-dependent reporter mice included the following: dual-dependent tdTomato line (Ai65; Jackson Laboratory #021875), dual-dependent ReaChR-mCitrine fusion protein line (Rosa26 CAG-LoxStopLox-FrtStopFrt-ReaChR-mCitrine; Jackson Laboratory #024846), dual-dependent GCaMP7s line (Ai195; Jackson Laboratory #034112); and dual-dependent mitochondrial matrix-targeted dimeric APEX2 line (ROSA26^{DR-Matrix-}dAPEX2; Jackson Laboratory #032764).

Mice were bred to have appropriate combinations of Cre, Flpo and reporter alleles. In the cases of KOR-Cre and *Bhlhe22*-Flpo alleles, all mice studied were heterozygous for each transgenic allele and therefore had one intact copy of each gene. Pilot studies that crossed either KOR-Cre with a Cre-dependent reporter line (Ai32) or *Bhlhe22*-Flpo mice with a Flp-dependent reporter line (Cai, Kardon, et al., 2016) suggested that both lines contained WACs with straight, unbranched dendrites, which motivated the intersectional approach described above. All mice were maintained on a 12/12-hour light/dark cycle and were studied at 1.5-6 months age. We did not expect or detect differences between male and female mice, and results have been combined across sex.

Electrophysiology and Cell Filling

Prior to retinal dissection, a mouse was dark-adapted for ~1 h. After death, both eyes were enucleated and maintained in Ames medium (MilliporeSigma) supplemented with 22.6 mM NaHCO_3 (MilliporeSigma) and suffused with 95% O_2 /5% CO_2 . The dissection was performed under infrared illumination using stereomicroscope-mounted night-vision goggles (OWL Night Vision Scopes, third generation; B. E. Meyers). The retina was removed from the eyecup, the vitreous humor was removed, and one or more relaxing cuts were made in order to flatten the retina. The

retina was mounted on a filter membrane (HAWP01300, MilliporeSigma) with five 0.5-mm diameter holes punched out and maintained in a dissection dish at room temperature for up to several hours. For recording, the retina was placed into a custom recording chamber and secured with a tissue harp. The recording chamber was perfused with Ames medium flowing at 4–6 mL min⁻¹ and maintained at 31 – 33 deg C.

Whole-cell patch-clamp recordings were obtained using patch pipettes pulled from borosilicate glass capillaries (1B120F-4, World Precision Instruments). Pipette tip resistances were ~4–10 MΩ for amacrine cell recordings in the ganglion cell layer; ~7–10 MΩ for amacrine cell recordings in the inner nuclear layer; and 2–10 MΩ for ganglion cell recordings. Pipettes were filled with internal solution containing (in mM): 120 K-methanesulfonate, 10 HEPES, 0.1 EGTA, 5 NaCl, 4 ATP-Mg, 0.4 GTP-Na₂, and 10 phosphocreatine-Tris₂, pH 7.3, 280 mOsm. The internal solution also contained 0.05% (w/v) Lucifer Yellow to label cells for subsequent immunohistochemical experiments.

The components of the internal solution were obtained from MilliporeSigma. Membrane potential was amplified (MultiClamp 700B, Molecular Devices), digitized at 10 kHz (Digidata 1440A, Molecular Devices) and recorded (pClamp 10.0, Molecular Devices). Series resistance ~20 – 40 MΩ for amacrine cell recordings in the GCL; ~20 – 100 MΩ for amacrine cell recordings in the INL; and ~8–25 MΩ for ganglion cell recordings. Series resistance was typically compensated by ~50%. Recordings were corrected for a –9 mV liquid junction potential.

Two-photon imaging

Two-photon fluorescent measurements were made using a custom-built microscope (Olympus BX-51) equipped with an Olympus 60×, 0.9 NA, LUMPlanFI/IR water immersion objective (Olympus), two fluorescence detection channels for GCaMP7s (HQ 535/50, Chroma) and SR101 (DM 565, 650/SP, Semrock), and an ultrafast pulsed laser (Chameleon Ultra II; Coherent) tuned to 910 nm. ScanImage software version 3.8.1 was used for image acquisition. Time-lapsed images (128 × 128 pixels; 26.8 × 26.8 μm frame) at each IPL depth were acquired at 7.8 frames per second with imaging laser power less than 30 mW. For volumetric imaging in the IPL, 12 time-lapsed z-stacks with 4-μm spacing were captured to cover the B/K WAC dendrites in the IPL. The z plane on the distal side (i.e., towards the INL) of the dendrites in the ON layer was defined as plane 0 for aligning across z stacks.

Visual stimulation

Visual stimuli were generated using custom software written in either Stim-Demo (C language) (Borghuis et al., 2014 [DOI](#); Borghuis et al., 2013 [DOI](#)) or in MATLAB using the Psychophysics Toolbox (Brainard, 1997 [DOI](#)) and were displayed by a DLP projector (HP Notebook Projection Companion) with 60-Hz refresh rate modified to project UV light (peak wavelength at 395 nm; Borghuis et al., 2013 [DOI](#)). This wavelength about equally activates cone photoreceptors along the retina's dorsal/ventral axis, which have varying ratios of green- and UV-sensitive opsins (Wang et al., 2011 [DOI](#)). The image of the DLP projector was focused on the retina through the microscope condenser.

Stimuli were presented within a 4 x 3 mm area on the retina and included contrast-reversing spots of variable diameter and drifting sine-wave gratings, both at 100% Michelson contrast relative to the background mean luminance that evoked ~10⁴ photoisomerizations cone⁻¹ s⁻¹. For the drifting grating stimuli used for calcium imaging, a full-field background of mean luminance was presented for 5 s to let the retina adapt to the combination of laser and mean luminance, and subsequently sinusoidal gratings moving in eight directions were presented sequentially. Each grating stimulus was presented for 2 s with a 5 s interval between stimuli. The grating was presented in a patch of 880 μm (29.3 degrees) in diameter and had a 2-Hz temporal frequency and 0.139 cycles/degree (4.54 cycles/mm) spatial frequency.

Spot tuning experiments were performed with spots of diameter 50 – 1675 μm . Because slightly different spot sizes were presented across experiments, responses were binned based on spot diameters centered around 100 μm (50 – 275 μm), 450 μm (276 – 625 μm), 800 μm (626 – 975 μm), 1150 μm (976 – 1325 μm) and 1500 μm (1326 – 1675 μm) before averaging across cells.

Imaging data analysis

Calcium imaging data were analyzed using custom software written in MATLAB. Data were analyzed within regions of interest (ROIs). To select ROIs, a Gaussian-blurred mean image was first subtracted from each frame for background fluorescence subtraction. Then, a local correlation image was calculated. Each pixel in the local correlation image was the mean of the correlation between the center pixel and the neighboring pixels in the surrounding 3×3 matrix. Rectangular ROIs were manually selected from non-overlapping dendritic segments on the correlated image, with one ROI per isolated dendritic segment. The ROIs were drawn as rectangles with the long side of the rectangle parallel to the dendrite. Next, the raw fluorescence and physical orientation θ_{phy} of each ROI were extracted based on the binary ROI mask. For the calcium response to the drifting gratings, $R(\theta_k)$ is calculated as the rectified mean of the baseline-subtracted fluorescence evoked by the grating moving in the direction θ_k for each ROI or image pixel. The baseline was defined as the average fluorescence 2.5 - 5 second after laser onset and just before stimulus onset.

An orientation selective index (OSI) was defined as follows:

$$OSI = \frac{|\sum_k R(\theta_k) e^{-2i\theta_k}|}{\sum_k R(\theta_k)}$$

which is equivalent to 1 – circular variance in circular statistics. A direction selective index (DSI) was defined as follows:

$$DSI = \frac{|\sum_k R(\theta_k) e^{-i\theta_k}|}{\sum_k R(\theta_k)}$$

The preferred orientation was calculated as follows:

$$\theta_{\text{pref}} = \frac{1}{2} \arctan\left(\frac{\sum_k R(\theta_k) \sin(2\theta_k)}{\sum_k R(\theta_k) \cos(2\theta_k)}\right)$$

The circular Pearson correlation (r_{circ}) between θ_{phy} and θ_{pref} in orientation space was defined as follows:

$$r_{\text{circ}} = \frac{\sum_j (\sin(2\theta_{\text{phy},j}) - \langle \sin(2\theta_{\text{phy}}) \rangle) (\sin(2\theta_{\text{ori},j}) - \langle \sin(2\theta_{\text{ori}}) \rangle)}{\sum_j (\sin(2\theta_{\text{phy},j}) - \langle \sin(2\theta_{\text{phy}}) \rangle) \sum_k (\sin(2\theta_{\text{ori},k}) - \langle \sin(2\theta_{\text{ori}}) \rangle)}$$

To demix spatially overlapping dendrites and visualize spatial components of apparent dendritic structures that were otherwise indistinguishable in the correlation image, we utilized non-negative matrix decomposition to decompose 3d video $Y(t)$ into products of spatial footprint A_i and corresponding temporal fluorescence traces C_i and background noise $B(t)$:

$$Y(t) = \sum_i A_i C_i(t) + B(t), \text{ s. t. } A_i \geq 0, \text{ and } C_i \geq 0$$

using the alternative least square algorithm (Berry et al., 2007 [DOI](#)).

Optogenetics

For optogenetic experiments, conventional photoreceptor-mediated input was pharmacologically blocked via bath application of the following drug cocktail (in μM): 50-100 D-AP5 (Alomone Labs), 50-100 DNQX (Alomone Labs), 20 L-AP4 (Alomone Labs) and 2-5 ACET (Tocris) (Park et al., 2015 [DOI](#); Park et al., 2020 [DOI](#); Park et al., 2018 [DOI](#); Pottackal et al., 2020 [DOI](#); Pottackal, Singer, & Demb, 2021 [DOI](#));

Pottackal, Walsh, et al., 2021). ReaChR-mediated responses were evoked by an LED (λ peak = 450 or 470 nm; maximum intensity $\sim 5 \times 10^{17}$ photons $s^{-1} cm^{-2}$) focused through the condenser onto a square (220 μm side) as described (Park et al., 2015 [↗](#); Park et al., 2020 [↗](#); Park et al., 2018 [↗](#)) and presented for 250-msec duration (average of 4 pulses, 1-s inter-pulse interval, repeated three times) The efficiency of ReaChR stimulation with blue light is within a factor of two as compared to the peak sensitivity using red light (Lin et al., 2013 [↗](#)). Optogenetically-evoked responses were measured as postsynaptic currents in RGCs. The response amplitude was the average current over 120 msec surrounding the peak response minus the baseline current, measured over 200 msec before light onset.

Histology

For whole-mount staining of filled cells, the retina was fixed after recording with 4% PFA for 1 h at room temperature. The retina was incubated in 6% donkey serum and 0.5% triton X-100 in PBS for 1 h at room temperature; and then incubated with 2% donkey serum and 0.5% triton X-100 in PBS with primary antibodies for 1 day at 4° C, and with secondary antibodies for 2 h at room temperature.

For staining with the Bhlhe22 antibody, the retina was fixed the tissue in 4% paraformaldehyde in PBS for 1 hour at room temperature; incubated in blocking solution overnight at 4°C; in primary antibody solution for 5 days at 4°C; and in secondary solution for 2 days at 4°C. Retinas were washed three times in PBS for 20 minutes each at room temperature after primary and secondary antibodies were applied. For imaging, the retinas were mounted on slides in Fluoromount G (Electron Microscopy Sciences) with the retinal ganglion cell layer facing the cover slip.

For staining retinal sections, eyecups were placed in 4% paraformaldehyde in PBS for 1-2 hours at room temperature. Next, eyecups were rinsed twice in PBS and embedded in 7% low-melt agarose (Precisionary, SKU VF-AGT-VM). The agarose-embedded eyecups were Vibratome-sectioned into 100 μm -thick slices (VT1200; Leica) and blocked in PBS containing 5% donkey serum and 0.5% Triton X-100 for 1 hour at room temperature. Sections were incubated with primary antibodies overnight at 4°C, washed three times with PBS for 10 minutes, and incubated for 2 hours at room temperature with appropriate secondary antibodies. Stained slices were washed three times in PBS for 10 minutes, mounted onto slides in Fluoromount G (Electron Microscopy Sciences) and cover slipped.

Primary antibodies were used at the following concentrations: rat anti-Bhlhe22 (Bhlhb5; 1:2000) (Ross et al., 2010 [↗](#)), goat anti-ChAT (1:200, Millipore AB144P, RRID:AB_2079751, rabbit anti-Lucifer Yellow (1:2000, ThermoFisher Scientific A-5750, RRID:AB_2536190), mouse anti-calretinin (1:5000, Millipore MAB1568). Secondary antibodies were conjugated to Alexa Fluor 488, C3 and C5 (Jackson ImmunoResearch or ThermoFisher Scientific) and diluted at 1:500.

Confocal imaging and analysis

For both retinal sections and flat mounts, confocal images were taken with a Zeiss LSM800 laser scanning confocal microscope. For filled cells, a whole-mount image of the dendritic tree was acquired using a 10X (NA = 0.3) or 20X air objective (NA = 0.8). A high-resolution z-stack was obtained to determine the relative depth of the filled dendrites relative to the ChAT bands, using a 40X oil objective (NA = 1.4); a digital zoom of 0.8 was used to expand the imaged area. Custom software written in MATLAB (MathWorks) was used to determine dendrite stratification relative to the ChAT bands.

The program and methods used were similar to those described previously (Beaudoin et al., 2019 [↗](#); Manookin et al., 2008 [↗](#); Park et al., 2015 [↗](#)). The depth of labeled processes is reported relative to the fluorescence peaks of the ChAT bands, in the z dimension, aligned to 0 (peak of inner ChAT band) and 1 (peak of outer ChAT band) in normalized units.

For the antibody staining in retinal flat mounts and sections, the 40X oil objective was used, and image analysis and processing was performed with *ImageJ* software (Fiji: ImageJ2 Version 2.3.0/1.53q).

Connectomic analysis with Scanning Block Face Electron Microscopy

Reconstructions were performed in an existing volume k0725 (mouse retina; 50×210×260 μm; voxel size 13.2×13.2×26 nm) (Ding et al., 2016 [DOI](#)); and with a new SBEM volume with labeled B/K WAC processes (61×24×40 μm; voxel size 5×5×40 nm). Manual skeletonization and annotation were performed using Knossos (Max Planck Institute for Medical Research; Heidelberg, Germany) (Helmstaedter et al., 2011 [DOI](#)). For analyses using k0725, voxel coordinates were tilt-corrected and normalized to the positions of the ON and OFF Starburst amacrine cells. Connectivity analysis was performed using custom-written Python scripts, and data were visualized using Paraview (Kitware, Inc.) (Graydon et al., 2018 [DOI](#); Grimes et al., 2022 [DOI](#); Park et al., 2020 [DOI](#)).

To prepare the new SBEM volume with labeled B/K WAC processes, a *Bhlhe22*-Flpo;KOR-Cre; dual-dAPEX2 mouse was euthanized and the retina was immersed for 30 minutes in 0.2M cacodylate buffer (pH 7.4) containing 4% paraformaldehyde, 2% glutaraldehyde, and 4 mM CaCl₂. Tissue was washed with 0.2M cacodylate buffer on ice (4×20 min), immersed for 30 min in 0.2 M cacodylate buffer containing 20 mM glycine on ice, and washed with 0.1M cacodylate buffer (pH 7.4) on ice (5×10 min). To visualize APEX2 expression, tissue was immersed for 30 minutes in 0.2 M cacodylate containing 0.5 mg/ml diaminobenzidine (DAB) on ice and protected from light. H₂O₂ was added to a final concentration of 0.03% and tissue was incubated for 20 minutes. Finally, tissue was washed in 0.2 M cacodylate buffer (pH 7.4) on ice (3×1 minutes, then 2×20 minutes). To prepare the tissue for SBEM, the sample was post-fixed at room temperature with 2% osmium tetroxide in 0.15 M cacodylate containing 2.5% potassium ferrocyanide, washed in three changes of distilled H₂O, incubated at 60° C in 1% aqueous thiocarbonylhydrazide, washed again in of distilled H₂O, incubated in distilled H₂O containing 1% aqueous uranyl acetate and 25% ethanol, rinsed in distilled H₂O, incubated in lead aspartate at 60° C, and rinsed in distilled H₂O. Then, dehydration was performed in a graded series of ethanol (35–100%) and infiltration in a propylene oxide/epoxy resin series was followed by embedding and polymerization in epoxy resin.

Transcriptomic analysis

Transcriptomic data came from a published study of single-cell RNA sequencing of amacrine cells (Yan et al., 2020 [DOI](#)). The database, available from the website of the Broad Institute (https://singlecell.broadinstitute.org/single_cell [DOI](#)), was downloaded and analyzed in custom software written in Matlab. The distribution of *Bhlhe22* expression was analyzed for each of the 63 previously assigned amacrine cell groups (i.e., presumed cell types).

Quantification and Statistical Analysis

Data are reported as mean ± SEM. P-values were calculated based on a two-tailed Student's T-test (paired or unpaired as appropriate). No statistical methods were used to predetermine sample sizes.

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Author contributions

Conceived the research plan: J.B.D., S.E.R., J.H.S.; Designed the experiments: S.J.P., W.L., J.P., S.E.R., J.H.S., J.B.D.; Collected and analyzed the data: S.J.P., W.L., J.P., A.O., J.M., J.P., C.H.-G., T.E.Z., C.C.-P., A.P.; Wrote the paper, original draft: J.B.D., W.L., J.H.S.; Edited the paper: T.E.Z., S.E.R., S.J.P.

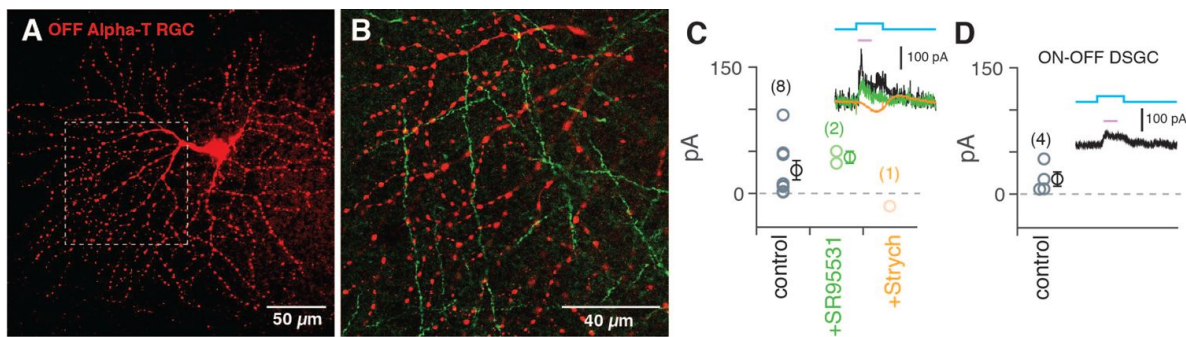
Declaration of Interests

The authors declare no competing interesting.

Resource Availability

Further information and requests for resources and reagents should be directly to Jonathan Demb (email: jonathan.demb@yale.edu).

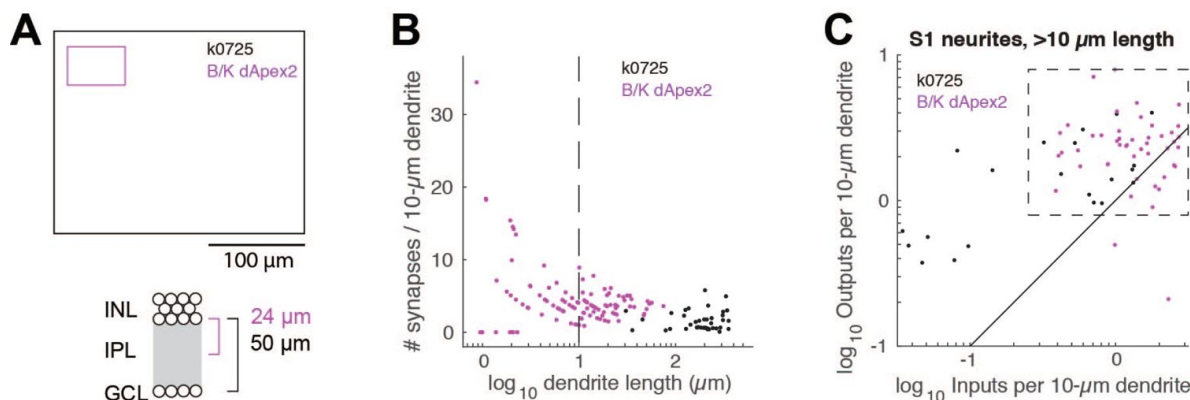
Supplementary Figure Legends



Supplementary Figure 1.

B/K amacrine cells make weak connections with certain retinal ganglion cell types

A. Dendritic tree of a recorded OFF Alpha RGC (also known as a Transient OFF Alpha RGC). Image shows average fluorescence in a confocal stack. Dashed square shows region in B. The RGC was filled with Lucifer Yellow (LY) during whole-cell recording, which was subsequently amplified with LY primary antibody and a red secondary antibody. **B.** Single confocal section showing OFF Alpha RGC dendritic tree relative to B/K WACs (Cre/Flpo-dependent ReaChR-mCitrine reporter). **C.** OFF Alpha RGCs had weaker IPSCs relative to OFF Delta and ON Alpha RGCs (Fig. 5). Responses were only weakly blocked by gabazine; in one cell, there was an inward current following addition of strychnine (inset). **D.** ON-OFF Direction-Selective RGCs showed weak IPSCs following optogenetic stimulation of B/K cells.



Supplementary Figure 2.

Studying labeled B/K amacrine cells with a scanning block face electron microscopy (SBEM) data set

A. Relative size and depth of the two SBEM datasets: an existing large dataset (k0725) and a smaller new data set with labeled B/K processes (B/K dApex2). The B/K dApex2 dataset focused on the OFF layer. **B.** There was an apparent bias to observe higher synapse density in short segments in the new dataset. We analyzed segments greater than 10 μm in length. **C.** Neurites in layer S1 had similar densities of input and output synapses in the new data set (B/K dApex2) and the subset of straight dendrites with dense synapses in the existing data set (k0725; boxed region).

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Editors

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Reviewer #1 (Public Review):

Summary:

This manuscript from Park et al examines the molecular, anatomical and functional properties of a subset of wide-field amacrine cell (WAC) types in mouse retina. More than 60 mouse amacrine cell types have been identified by single-cell transcriptomic studies (Yan et al., 2020, PMID: 32457074), but the functions of most of these are unknown and WACs are particularly understudied. The authors use intersectional genetics to target a subset of mouse

WACs that co-express Bhlhe22 and the kappa opioid receptor (referred to as B/K WACs). They used electrophysiological and anatomical approaches to determine how WACs contribute to neural computations in the retina.

Strengths:

Overall, the paper presents a technically impressive set of experiments that build strong evidence for the presence of at least 3 discrete WAC types in the B/K transgenic line. These cells vary with respect to their morphology, dendritic stratification, response polarity (On vs Off) and resting membrane potentials. All types have long, monostatified dendrites and appear to lack axons. Electrophysiological recordings establish that these WACs are non-spiking, while calcium imaging revealed orientation selectivity in dendritic segments with tuning that correlates strongly with dendritic orientation. The authors go on to use optogenetics to show that WACs provide strong GABA-A receptor mediated inhibitory input to OFF and ON alpha sustained RGCs. This connectivity is further substantiated, at least for the OFF sustained alpha RGCs, by connectomic analyses from serial block face EM volumes. The use of the APEX2 reporter system to label the B/K cells in one of the EM volumes is particularly nice, making identification of the B/K WACs unambiguous. The conclusions are largely well supported by the experimental data. The study provides novel insights into the structure and function of specific WACs that will provide a foundation for further studies investigating the role of these amacrine cells in retinal circuits.

Weaknesses:

A limitation of the study is that the B/K WAC types described here could not be aligned to specific transcriptomic identities. The authors show more than 15 GABA expressing ACs express Bhlhe22 in the transcriptomic dataset, but it is unclear which of these also express the kappa opioid receptor (Opkr1).

The optogenetic evidence suggests that WACs provide GABA-A receptor mediated inhibitory input to both the sustained OFF and ON alpha RGCs. However, at least in the examples shown, there appears to be a dramatic difference in the timecourse of the rising phase of the inhibitory inputs to these two cell types, with the inputs to the ON sustained alpha RGCs appearing slower than those in the OFF sustained and OFF transient alpha RGCs. This apparent temporal difference was accompanied by a relatively lower sensitivity to light stimulation for the ON sustained cells. The slow timecourse seems unexpected for a direct GABA-A mediated synaptic connections between the WACs and ON alpha sustained cells. Moreover, since the connectomic analyses do not examine inputs to ON RGC types, the direct synaptic connection between B/K WACs and On alpha RGC is less well substantiated.

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Reviewer #2 (Public Review):

Summary:

An important frontier in research on the mammalian retina is to understand the role of inhibitory amacrine cells in visual processing. These cell types have been found to play roles in tuning the output of the retina to specific visual features like motion and orientation. These cell types are understudied for two main reasons. First, there are many types of them—over 60 types in the mouse—, and second, they are quite unconventional as far as neurons go, as they have dendrites but often lack axons. The manuscript "Molecular identification of wide-field amacrine cells in mouse retina that encode stimulus orientation" by Park et al. provides a characterization of two (or possibly more) cell types within the amacrine cell class. Specifically, they characterize types of widefield amacrine cells (WACs), which they have

gained genetic access to using an intersectional transgenic mouse strategy (Bhlhe22 x KOR). The authors used a broad range of experiments to characterize these WACs' anatomical properties, their stimulus tuning, and their wiring within the retina to their postsynaptic partners. These experiments include anatomy, electrophysiology, calcium imaging, and electron microscopy.

Strengths:

Overall, the manuscript presents strong evidence that the Bhlhe22 x KOR WACs represent multiple WAC types in the retina and that these cell types are orientation tuned. The most exciting finding is that their orientation tuning is correlated with the physical orientations of the dendrites, which suggests that this anatomical feature supports the tuning in these cells.

Weaknesses:

(1) The one common thought about widefield amacrine cells (WACs) is that these are spiking cells, which allows them to transmit signals along their long dendrites. The authors state that "none of the recorded cells fired conventional action potentials (spikes)." (p.7) This is a surprising finding, which leads to an interesting question: how do these cells integrate information from their presynaptic partners to generate the orientation tuning observed without the ability to conduct over long distances? However, the authors have not fully ruled out that the cells do spike.

For instance, one possibility is that spiking requires a specific stimulus and the authors did not play that stimulus during their recordings. Most somatic recordings did not result in very large depolarizations, and the cell could still be below threshold. Depolarizing the cell to attempt to evoke spikes directly could be used to explore this possibility. A second possibility is that the dendrites spike, but these spikes are attenuated at the soma. Direct injections of current into the cells to evoke such spikes could be used to observe whether dendritic spiking occurs. A third possibility is that some important machinery for spiking is being washed out by the whole cell recordings. Cell attached recordings could be used to assess whether spiking occurs in an intact cell. The authors may wish to address these possibilities experimentally, but at least should qualify their statement about spiking in these cells and discuss these possibilities.

(2) It was unclear in this paper how many cell types are present in the intersectional cross. I think the paper would be stronger if they clarified that. For instance, in Fig. 1B: the authors show Bhlhe22 expression in amacrine cells from a previous study. They should also show the expression of the other gene they used in their intersectional strategy, the Kappa Opioid receptor (Oprk1), which is available in the same dataset. Another piece of analysis that could help would be clearer quantification of the anatomical features of the cells. For instance, the cells shown in Fig. 2 A2 vs. B2 have clear differences in number of dendrites and the relative angles of the dendrites. The On cells appear to have more dendrites evenly spread around the soma, while the Off cells appear to have more clumping along a line. Is this the case for all the cells recorded, or just these examples? The authors should present some population-level quantification.

(3) In Fig. 4E, the preferred orientation of calcium responses and physical orientation of the dendrites appears to clump around specific orientations. The Methods don't mention if the retinas were aligned to the body axis during the dissection. Is this clumping real, or is this an artifact of the analysis? If there are specific preferred orientations to these WAC cell types, that would be important to discuss in the paper - for instance how this relates to the preferred direction in the direction selectivity system or how it might relate to the function of these cells for behavior.

<https://doi.org/10.7554/eLife.94985.1.sa1>

Reviewer #3 (Public Review):

Summary:

Amacrine cells are a heterogeneous collection of retinal interneurons. Most are inhibitory, and like inhibitory neurons in other neural circuits, strongly shape retinal function. With a few exceptions, the role of amacrine cells in retinal signaling is poorly understood. This paper introduces an approach to study a set of wide-field amacrine cells that extend processes over large regions of the retina.

Strengths:

A substantial strength of the paper is the combination of genetic manipulations, electrophysiology, optogenetics and electron microscopy used to study these cells. As a result of that broad set of techniques, the results cover many properties of how the cells work and provide a nice overview. The paper is also (with a few exceptions below) clearly presented and the experiments look to be carefully executed with clean results.

Weaknesses:

My largest concern with the paper is that overall the results provided an initial view of an interesting set of issues about the function of these cells, but the interesting initial results are not pursued in more depth.

Spatial spread of signals in neurites

An immediate question about axonless WACs is the extent of spread of signals along their processes, and hence whether they act as a collection of independent or semi-independent elements. This bears directly on interpretation of the responses to oriented stimuli for example. Did you do any experiments that might provide additional information about this issue? For example, if you stimulate one of the WACs peripherally do you see a strong modulation of the somatic voltage? Or in the imaging experiments, if you mask a region of the processes so that it is not receiving a stimulus, do you see responses "leak" into that occluded region from surrounding stimulated regions?

Orientation tuning and connectivity

The most developed functional results in the paper relate to the sensitivity of the WAC processes to oriented stimuli. Interpretation of these results depends on a few factors. First is the spread of signals in the WAC processes - as noted above. Second is connectivity. The paper shows that the B/K WAC activity increases inhibitory input to Off-delta and On-alpha ganglion cells. These cells, as noted in the paper, are not orientation tuned. But the orientation tuned ganglion cells stratify in a similar location within the IPL, and hence are situated in an appropriate place to receive input from the B/K WACs. Did you focus exclusively on connections to the Off-delta and On-alpha cells (along with the Off-alpha) or did you look at any other ganglion cell types? This should at least get discussed in more detail.

In several places it is unclear whether the paper intends to be a methods paper or a basic research paper. One example is the last sentence of the abstract. If it is intended to be a basic research paper (which is my overall impression) then I suggest shifting the emphasis in some of those key locations towards results and away from methods.

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